

Comprehensive Security Exam Study Guide

This guide is aligned to the provided sample exam style questions. True/False traps, all correct multi-select logic, short mechanical explanations, and concise risk matrices. For every attack/vulnerability concept, the guide explicitly tracks:

- Type of Attack
- Threat Classification, usually STRIDE
- Security Goal(s) Violated: Confidentiality, Integrity, Availability, Authenticity, Authorization, Accountability, Accuracy, Non-repudiation, and Privacy where relevant

Course-Wide Exam Anchors

- Security Goal(s) Violated: Confidentiality, Integrity, Availability, Accuracy
- Confidentiality: only authorized parties can access information, violated by grade leakage, cookie theft, Heartbleed memory disclosure, tracking leaks, database dumping, DNS redirection to malicious endpoints that exposes data
- Integrity: information or system state is not improperly modified. Violated by SQL injection, DNS poisoning, tampering with hidden form values, buffer overflow that changes program variables, unauthorized code execution
- Availability: authorized users can access information/resources reliably. Violated by DoS, Denial of Service (DoS), Distributed Denial of Service (DDoS), ransomware
- Authenticity: an entity, message, or source is genuine and verifiable. Violated by spoofing, MITM, unauthenticated DNS, email spoofing, fake DNS certificates
- Authorization: unauthorized users cannot access information or resources. Violated by session hijacking, impersonation, privilege escalation
- Accountability: actions are uniquely tied to the user. Violated by missing logs, shared credentials, spoofing that prevents attribution, reputation attacks
- Non-repudiation: a party cannot later deny an action or message. Violated by lack of digital signatures, lack of logs, tampering with evidence
- Accuracy: information is suitable/correct for its purpose. Violated when data is wrong even if maliciously tampered, or people corrupted grades or test scores
- Privacy: personal data is collected and processed according to expectations/consent. Violated by online tracking, session replay leaks, browser fingerprinting, location-data inference, doxxing/stalking

STRIDE Mapping

STRIDE Spoofing	Can someone else pretend to be you?	Can someone change data or code?	Can you prove who did it?	Can data leak?	Can access be prevented?	Can a loss of privilege/gain higher privilege occur?
Tampering	Confidentiality, Integrity, Availability, Accuracy	Integrity, Accuracy, Confidentiality, Availability	Confidentiality, Availability	Confidentiality, Privacy	Availability	Availability
Reputation	Confidentiality, Availability	Confidentiality, Availability	Confidentiality, Availability	Confidentiality, Privacy	Availability	Availability
Information disclosure	Confidentiality, Availability	Confidentiality, Availability	Confidentiality, Availability	Confidentiality, Privacy	Availability	Availability
Denial of service	Confidentiality, Availability	Confidentiality, Availability	Confidentiality, Availability	Confidentiality, Privacy	Availability	Availability
Elevation of privilege	Confidentiality, Availability	Confidentiality, Availability	Confidentiality, Availability	Confidentiality, Privacy	Availability	Availability

Week 1: Introduction to Information Security

1. Core Technical Terms & Definitions

- Computer security: the practice of preserving confidentiality, integrity, and availability of system assets: hardware, software, firmware, and processed data/transmitted information
- Information Systems Security (INFOSEC): Security: protection of information systems against unauthorized access, modification, and denial of service, including detection, deterrence, and counteraction
- Information Security: protection of information and systems from unauthorized access, use, disclosure, disruption, modification, or destruction to provide CIA
- CIA triad: Confidentiality, Integrity, Availability
- Additional AAAA-style properties: Authenticity, Authorization, Accountability, Availability
- Non-repudiation: proof of origin/validity/sequence such that parties cannot deny involvement later
- Asset: something restricting access to use of personal or relying-party information according to law/policy
- Asset: something valuable to protect, such as systems, data, people, equipment, information, reputation, or trustworthiness
- Vulnerability: a weakness in a system, process, control, application, or network that could be exploited
- Threat: a possible adverse circumstance/event or threat source that can exploit a vulnerability
- Exploit: an action that causes a vulnerability to be exploited
- Risk: function of likelihood and impact, given a threat exploiting a vulnerability against an asset
- Countermeasure: a control: safeguard deployed to reduce risk
- Control: a component: a component whose properties the system relies on
- Trustworthy component: a component that deserves trust; not all trusted components are trustworthy
- Trusted Computing Base (TCB): minimal set of components that must behave correctly for security
- Security control types by nature: physical, logical/technical, administrative/policy
- Security control types by timing: preventive/deterrent, detective, corrective/responsive

- Unauthorized data access
 - Type of Attack: Information exposure through broken access restrictions
 - Mechanical breakdown:
 - 1. An asset contains sensitive data, such as grades, personal records, or proprietary data
 - 2. The system exposes the data to users who do not have a valid need or privilege
 - Threat Classification: STRIDE Information Disclosure; outside their Authorization of Privilege if it is not intended
 - Security Goal(s) Violated:
 - Confidentiality: unauthorized users can see protected information
 - Authorization: access was granted to users who should not have it
 - Privacy: if the data is personal data
- Unauthorized modification
 - Type of Attacks: Tampering with data or system state
 - Mechanical breakdown:
 - 1. Attacker obtains a path to modify records, files, configuration, or code
 - 2. The modification changes correctness or trustworthiness
 - 3. Downstream decisions rely on corrupted data
 - 4. Draw trust boundaries
 - Threat Classification: STRIDE Tampering
 - Security Goal(s) Violated:
 - Integrity: data/system state is improperly changed
 - Accuracy: information may no longer be used for its purpose
 - Accountability: if the system cannot trace who changed it

- Service disruption
 - Type of Attack: Denial of Service
 - Mechanical breakdown:
 - 1. Attacker overloads, crashes, disables, or otherwise blocks a resource
 - 2. Legitimate users cannot access the resource
 - 3. The organization suffers operational, reputational, or financial impact
 - Threat Classification: STRIDE Denial of Service
 - Security Goal(s) Violated:
 - Availability: authorized users are denied timely and reliable access

- Spoofed identity or origin
 - Type of Attack: Spoofing / impersonation
 - Mechanical breakdown:
 - 1. Attacker claims to be a trusted user, process, host, sender, or service
 - 2. The system accepts the false identity due to weak or missing authentication
 - 3. The attacker gains trust, receives data, or performs actions
 - Threat Classification: STRIDE Spoofing
 - Security Goal(s) Violated:
 - Authenticity: claimed originator is not genuine
 - Authorization: access is granted to users who should not have it
 - Accountability: attribution may point to the wrong entity

- 3. Conceptual Frameworks & Defensive Distinctions
 - Asymmetric cryptography
 - Exploitation creates impact; likelihood and impact determine risk; controls reduce risk
 - Control types: locks, guards
 - Physical detective: cameras
 - Physical corrective: sprinklers
 - Technical preventive: firewall, access control, encryption
 - Technical detective: IDS, audit logging
 - Technical corrective: backups, restore procedures
 - Administrative preventive: security awareness training, security screening policy
 - Administrative detective: log review policy
 - Administrative corrective: incident response and legal liaison
 - Security is a process, not a target: systems change, threats change, and controls require monitoring and adaptation
 - Important exception: Integrity in the CIA domain excludes authenticity and non-repudiation in some formal definitions, but exam multi-select questions often ask for the most direct violated goals. Mention the specific property when it is clearly present

Week 2: Risk Management and Threat Modeling

1. Core Technical Terms & Definitions

- Risk management: discipline of identifying, evaluating, treating, and monitoring risks
- Risk assessment / risk analysis: evaluating assets, threats, vulnerabilities, likelihood, and impact, and control
- Risk formula: risk = likelihood x impact
- Qualitative risk assessment: subjective, based on experience, such as low/moderate/high, often used when precise measurement is impractical
- Quantitative risk assessment: numerical estimation of impact, likelihood, expected loss, and control
- Risk exposure: expected cost of incidents before or after controls
- Risk reduction: amount of risk removed by a control
- Risk avoidance: risk reduction benefit compared with control cost
- ROI: Return on Security Investment, commonly (risk reduction - control cost) / cost
- Risk avoidance: stop the risky activity or remove exposure
- Risk transfer: shift risk financially/contractually, for example insurance or outsourcing, while remembering ultimate liability often remains with controls
- Risk mitigation: reduce likelihood or impact with controls
- Risk acceptance / assignment: knowingly accept residual risk and design ownership
- Countermeasure selection: choosing controls based on cost, effectiveness, business impact, residual risk, and fit to the threat
- ISO 31000: risk management standard
- NIST SP 800-30: guide for conducting risk assessments
- NIST SP 800-35: catalog of security and privacy controls
- Threat modeling: structured analysis of potential threats before or during design
- CRAMM, OCTAVE, FRA, FMEA, FTA, VAR: examples of risk methods with different qualitative/quantitative focus
- Data Flow Diagram (DFD): model of actions, processes, data stores, and data flows
- Trust boundary: border between zones with different trust assumptions, threats often arise when data crosses it
- STRIDE: Spoofing, Tampering, Repudiation, Information disclosure, Denial of Service, Elevation of privilege
- Separation of Duties (SoD): design principle splitting sensitive responsibilities across multiple people
- Least privilege: users/processes should have only the privileges needed for their task
- Fail-safe defaults: deny by default unless access/right is explicitly allowed
- Defense in depth: multiple independent layers of protection
- Security Goal(s) Violated:
 - Authenticity: identity verification fails
 - Authorization: legitimacy verification fails
 - Accountability: logs may attribute actions to the wrong identity

- Tampering in STRIDE
 - Type of Attack: Unauthorized modification of data, code, messages, or configuration
 - Mechanical breakdown:
 - 1. Attacker obtains write influence over a data flow, data store, file, request, or command
 - 2. The value is changed without authorization
 - 3. The system controls the alteration
 - Threat Classification: STRIDE Tampering
 - Security Goal(s) Violated:
 - Integrity: altered content shares expected dignity
 - Authenticity: if hash or other evidence is not present, it is not trusted
 - Authorization: if write privileges were bypassed

- Information disclosure in STRIDE
 - Type of Attack: Denial of confidentiality / impersonation
 - Mechanical breakdown:
 - 1. Attacker claims an identity, role, host, service, or session token
 - 2. The system lacks sufficient authentication/session validation
 - 3. The attacker is treated as legitimate
 - Threat Classification: STRIDE Spoofing
 - Security Goal(s) Violated:
 - Confidentiality: information is not genuine
 - Authenticity: identity verification fails
 - Accountability: logs may attribute actions to the wrong identity
- Reputation in STRIDE
 - Type of Attack: Denial of performed action due to insufficient proof
 - Mechanical breakdown:
 - 1. A user or process performs a sensitive action
 - 2. Logs, signatures, timestamps, or audit trails are missing/wrong
 - 3. The action can plausibly deny pasting it
 - Threat Classification: STRIDE Repudiation
 - Security Goal(s) Violated:
 - Accountability: action cannot be traced unambiguously
 - Non-repudiation: action can deny the action
- Information disclosure in STRIDE
 - Type of Attack: Data leakage / unauthorized read
 - Mechanical breakdown:
 - 1. Sensitive information exists in a data store, data flow, memory region, log, or UI
 - 2. Access control, encryption, or isolation fails
 - 3. Unauthorized parties learn the information
 - Threat Classification: STRIDE Information Disclosure
 - Security Goal(s) Violated:
 - Confidentiality: data is exposed
 - Privacy: personal data and behavioral data is exposed
- Denial of Service in STRIDE
 - Type of Attack: Resource exhaustion, crash, outage, request flood, dependency outage
 - Mechanical breakdown:
 - 1. Attacker consumes capacity or triggers failure
 - 2. The system cannot process legitimate requests
 - 3. Users/services experience degraded or unavailable access
 - Threat Classification: STRIDE Denial of Service
 - Security Goal(s) Violated:
 - Availability: access is blocked or unreliable

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 - Asymmetric cryptography
 - Exploitation with a symmetric key provides mutual protection
 - Hash functions without a secret key detect accidental/incidental changes only when the expected hash is trusted through a trusted channel
 - MACs use a secret key and provide data integrity plus data origin authentication between parties sharing the key
 - MAC exam trap: MACs are not cryptographic hash functions. HMAC is exactly a hash-based MAC
 - Non-repudiation trap: symmetric MACs do not provide strong non-repudiation between the two parties because they share the same key or either could have produced the MAC
 - Casual cipher decryption with key 3: shift ciphertext letters 3 positions backward, z becomes y
 - Authenticated encryption is needed when both confidentiality and integrity/authenticity of encryptions are required

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Week 7: Application Security II - Defenses

